SHIBANI SINGH

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SUMMARY

Machine Learning / Systems Engineer with 8+ years of experience building production AI systems, ML infrastructure, benchmarking workflows, and developer-facing automation tools. Strong background in AI workload evaluation, distributed ML pipelines, profiling, search/indexing optimization, and performance analysis across Intel and cloud environments. Recent experience designing LLM-powered multi-agent workflows for enterprise automation, including retrieval, planning, validation, monitoring, and human-in-the-loop execution. Skilled in Python, C++, PyTorch, TensorFlow, distributed systems, and scalable tooling for ML engineers.

EDUCATION

M.S. in Cybersecurity, New York University — 2023 – 2025
Recipient of NYU Cyber Fellows Scholarship — GPA: 3.8/4.0
CAE-aligned Cyber Defense / Cyber Operations coursework

M.S. in Computer Science, Arizona State University — 2015 – 2017
Eta Kappa Nu (IEEE Honors Society) — GPA: 3.8/4.0
Master's Thesis: Deep Learning-based Classification of FDG-PET Data for Alzheimer's Disease (NSF-funded research)

B.Tech in Computer Science and Engineering, The LNM Institute of Information Technology — 2011 – 2015
Thesis: Side-channel analysis of ECDSA in Bitcoin mining — GPA: 8.8/10.0

EXPERIENCE

Lead Machine Learning Engineer — Jul 2023 – Present
Platform Architecture & Automation, Hilton — *Remote, US*

- Designed and deployed LLM-powered multi-agent workflows to automate enterprise business processes across document understanding, invoice analysis, semantic retrieval, validation, routing, and downstream task execution.
- Built agent orchestration patterns involving planner, retriever, validator, anomaly-detection, and human-review agents, reducing manual review effort by 40%+ across workflows processing marketing tech. QA workflows, sales and finance workflows.
- Developed production LLM/RAG pipelines spanning ingestion, chunking, embeddings, vector indexing, ranking, context selection, evaluation, monitoring, and fallback handling on AWS.
- Created internal automation tools used by platform, product, and operations teams to translate user requirements into scalable workflows with traceability, observability, and quality controls.
- Improved LLM retrieval and automation quality through systematic prompt/version testing, ranking optimization, error analysis, latency monitoring, and production failure-mode tracking.
- Partnered with product, platform, security, and operations teams to deploy reliable AI workflows with measurable business impact, observability, and scalable engineering practices.

AI Applied Research Scientist — Oct 2020 – May 2023
Security & Privacy Research Group, Intel Labs — *Oregon, US*

- Conducted applied research on robustness and reliability of deep learning systems under DARPA GARD, evaluating model behavior under adversarial and distribution-shift scenarios.
- Designed experimental frameworks, threat models, and evaluation pipelines for real-world ML systems with safety, reliability, and performance constraints.
- Created an open-source CARLA-based synthetic data generation framework for ML evaluation and adversarial research, adopted by multiple research teams.

- Published peer-reviewed work at ICML workshops and collaborated with cross-functional teams bridging research, evaluation, and production-facing ML systems.

Cloud Solutions Engineer Jun 2017 – Oct 2020
Data Center & AI Group, Intel Corporation *Oregon, US*

- Built benchmarking and performance-analysis workflows for ML, analytics, and search workloads across Intel and cloud environments, delivering 2–4× performance gains through profiling and systems-level tuning.
- Identified workload bottlenecks across ingestion, indexing, distributed execution, and query paths, translated findings into reusable optimization patterns for engineering and customer deployments.
- Developed scalable data and ML pipelines for large-scale ingestion, pre-processing, model execution, and analytics using Spark, Hadoop, Kafka, Linux, and cloud infrastructure.
- Partnered with engineering teams to analyze workload performance, document tuning methods, and improve portability, efficiency, and production readiness across deployments.
- Optimized Splunk search and indexing performance on Intel Architecture and received two Intel Division Recognition Awards for analytics performance optimization.

Software Engineering Intern May 2016 – Aug 2016
Intel Corporation *Oregon, US*

- Developed ML algorithms for the Trusted Analytics Platform on AWS, improving distributed analytics throughput and model performance.

Software Engineering Intern Jan 2014 – May 2014
IIT Bombay, MOOC Lab *Mumbai, India*

- Built backend systems for large-scale e-learning platforms and automated grading pipelines presented work at LWMOOCs Workshop, MIT.

SKILLS

Languages	Python, C, C++, Scala
AI / Frameworks	PyTorch, TensorFlow, Scikit-learn, LLMs, RAG, Model Evaluation, Inference Pipelines
Systems / Performance	Benchmarking, Profiling, Workload Characterization, Distributed ML Optimization
Systems	Distributed Systems, Spark, Kafka, Hadoop, YARN, Docker, Linux, AWS
Data / Search	Vector Search, Retrieval Systems, PostgreSQL, MySQL
Automation	Multi-Agent Workflows, Workflow Orchestration, Human-in-the-Loop Systems

PUBLICATIONS & PRESENTATIONS

2022	Synthetic Data Generation for Adversarial ML Research – ICML AdvML
2019	High Performance Data Analytics with Splunk – Splunk .conf
2018	Deep Learning Opinion Mining – Spark AI Summit
2017	Deep Learning-based Classification of FDG-PET Data for Alzheimer's Disease – SPIE

AWARDS

2020	Intel Division Recognition Awards
2014	Full Funding Scholarship – MIT LWMOOCs Workshop